

ESE5180 Lab 0 Zephyr

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GitHub repository: <https://github.com/curious88sniper/ese5180-lab0.git>

1. Vanilla Zephyr Blinky

Video for this section is submitted separately through the course Google Form as required by the assignment.

2. nRF Connect SDK Blinky

Video for this section is submitted separately through the course Google Form as required by the assignment

3. Building With West

3.1 Build and Flash Screenshots

Build output:

```

PS D:\ncs\v3.4.0\zephyr> west build -p always -b nrf7002dk/nrf5340/cpuapp/ns -d D:\ncs\lab0-build-ns D:\ncs\lab0
-app --sysbuild
-- west build: making build dir D:\ncs\lab0-build-ns pristine
-- west build: generating a build system
Loading Zephyr module(s) (Zephyr base): sysbuild_default
-- Found Python3: D:/ncs/toolchains/dcbdc366a1/opt/bin/python.exe (found suitable version "3.12.4", minimum requ
ired is "3.12") found components: Interpreter
-- Cache files will be written to: D:/ncs/v3.4.0/zephyr/.cache
-- Found west (found suitable version "1.5.0", minimum required is "0.14.0")
-- Board: nrf7002dk, qualifiers: nrf5340/cpuapp/ns
Parsing D:/ncs/v3.4.0/zephyr/share/sysbuild/Kconfig
Loaded configuration 'D:/ncs/lab0-build-ns/_sysbuild/empty.conf'
Merged configuration 'D:/ncs/lab0-build-ns/_sysbuild/empty.conf'
Configuration saved to 'D:/ncs/lab0-build-ns/zephyr/.config'
Kconfig header saved to 'D:/ncs/lab0-build-ns/_sysbuild/autoconf.h'
--
*****
* Running CMake for lab0-app *
*****

Loading Zephyr default modules (Zephyr base).
-- Application: D:/ncs/lab0-app
-- CMake version: 4.2.1
-- Found Python3: D:/ncs/toolchains/dcbdc366a1/opt/bin/python.exe (found suitable version "3.12.4", minimum requ
ired is "3.12") found components: Interpreter
-- Cache files will be written to: D:/ncs/v3.4.0/zephyr/.cache
-- Zephyr version: 4.4.0 (D:/ncs/v3.4.0/zephyr)
-- Found west (found suitable version "1.5.0", minimum required is "0.14.0")
-- Board: nrf7002dk, qualifiers: nrf5340/cpuapp/ns
-- Found host-tools: zephyr 1.0.1 (D:/ncs/toolchains/dcbdc366a1/opt/zephyr-sdk)
-- Found toolchain: zephyr 1.0.1 (D:/ncs/toolchains/dcbdc366a1/opt/zephyr-sdk)
-- Found Dtc: D:/ncs/toolchains/dcbdc366a1/opt/bin/dtc.exe (found suitable version "1.4.7", minimum required is
"1.4.6")
-- Found BOARD.dts: D:/ncs/v3.4.0/zephyr/boards/nordic/nrf7002dk/nrf7002dk_nrf5340_cpuapp_ns.dts
-- Found devicetree overlay: D:/ncs/lab0-app/boards/nrf7002dk_nrf5340_cpuapp_ns.overlay
-- Generated zephyr.dts: D:/ncs/lab0-build-ns/lab0-app/zephyr/zephyr.dts
-- Generated pickled edt: D:/ncs/lab0-build-ns/lab0-app/zephyr/edt.pickle

```

Memory region	Used Size	Region Size	%age Used
FLASH:	81364 B	256 KB	31.04%
RAM:	45960 B	256 KB	17.53%

[195/195] Linking C static library secure_fw\libtfm_s_veneers.a

```

[17/259] Performing install step for 'tfm'
[17/259] Performing install step for 'tfm'
-- Install configuration: "MinSizeRel"
----- Installing platform NS -----
-- Install configuration: "MinSizeRel"
----- Installing platform NS -----
----- Installing platform NS -----
[259/259] Linking C executable zephyr\zephyr.elf
Memory region      Used Size  Region Size  %age Used
[259/259] Linking C executable zephyr\zephyr.elf
Memory region      Used Size  Region Size  %age Used
Memory region      Used Size  Region Size  %age Used
Memory region      Used Size  Region Size  %age Used
FLASH:             22576 B      704 KB      3.13%
FLASH:             22576 B      704 KB      3.13%
RAM:               4600 B       192 KB      2.34%
IDT_LIST:          0 B         32 KB       0.00%
Generating files from D:/ncs/lab0-build-ns/lab0-app/zephyr/zephyr.elf for board: nrf7002dk/nrf5340/cpuapp/ns

[8/8] Completed 'lab0-app'
PS D:\ncs\v3.4.0\zephyr>

```

Flash output:

```

[2/3] Completed 'lab0-app'
-- west flash: rebuilding
[0/16] Performing build step for 'tfm'ninja: no work to do.

[2/3] Performing install step for 'tfm'
-- Install configuration: "MinSizeRel"
----- Installing platform NS -----
[3/3] Completed 'tfm'

[3/3] Completed 'lab0-app'
-- west flash: using runner nrfutil
-- runners.nrfutil: reset after flashing requested
-- runners.nrfutil: Flashing file: D:\ncs\lab0-build-ns\lab0-app\zephyr\tfm_merged.hex
-- runners.nrfutil: Connecting to probe
-- runners.nrfutil: Erasing address ranges touched by firmware
-- runners.nrfutil: Programming image
-- runners.nrfutil: Verifying image
-- runners.nrfutil: Reset
-- runners.nrfutil: Board(s) with serial number(s) 1050773318 flashed successfully.
PS D:\ncs\v3.4.0\zephyr>

```

West Commands

- **west init**: Initializes a Zephyr workspace and configures the manifest repository.
- **west update**: Downloads and checks out the repositories and revisions listed in the manifest.
- **west build**: Configures and builds a Zephyr application through CMake and Ninja.
- **west flash**: Programs an already-built firmware image onto the selected board.

Zephyr wraps CMake with west because a Zephyr workspace is usually a multi-repository project. West gives one command-line tool for workspace management, build configuration, flashing, debugging, and extension commands.

Build Command

```

west build -p always -b nrf7002dk/nrf5340/cpuapp/ns -d D:\ncs\lab0-build-ns
D:\ncs\lab0-app --sysbuild

```

- **-p always**: Makes the build pristine before configuring, so stale build files are removed.
- **-b nrf7002dk/nrf5340/cpuapp/ns**: Selects the nRF7002 DK nRF5340 non-secure application core board target.

- `-d D:\ncs\lab0-build-ns`: Sets the output build directory.
- `D:\ncs\lab0-app`: Sets the application source directory.
- `--sysbuild`: Builds the application with Zephyr sysbuild, which is needed for this nRF Connect SDK target because TF-M secure firmware is built alongside the non-secure app.

Flash Command

```
west flash -d D:\ncs\lab0-build-ns --dev-id 1050773318
```

- `-d D:\ncs\lab0-build-ns`: Selects the existing build directory to flash.
- `--dev-id 1050773318`: Selects the connected nRF7002 DK debug probe by serial number.

4. Kconfig

Sourcing Zephyr in Kconfig is necessary because the application Kconfig file depends on Zephyr's base symbols, menus, defaults, and validation rules.

5. Devicetree Overlays

```
aliases {  
    led5180 = &led1;  
    button5180 = &button0;  
};
```

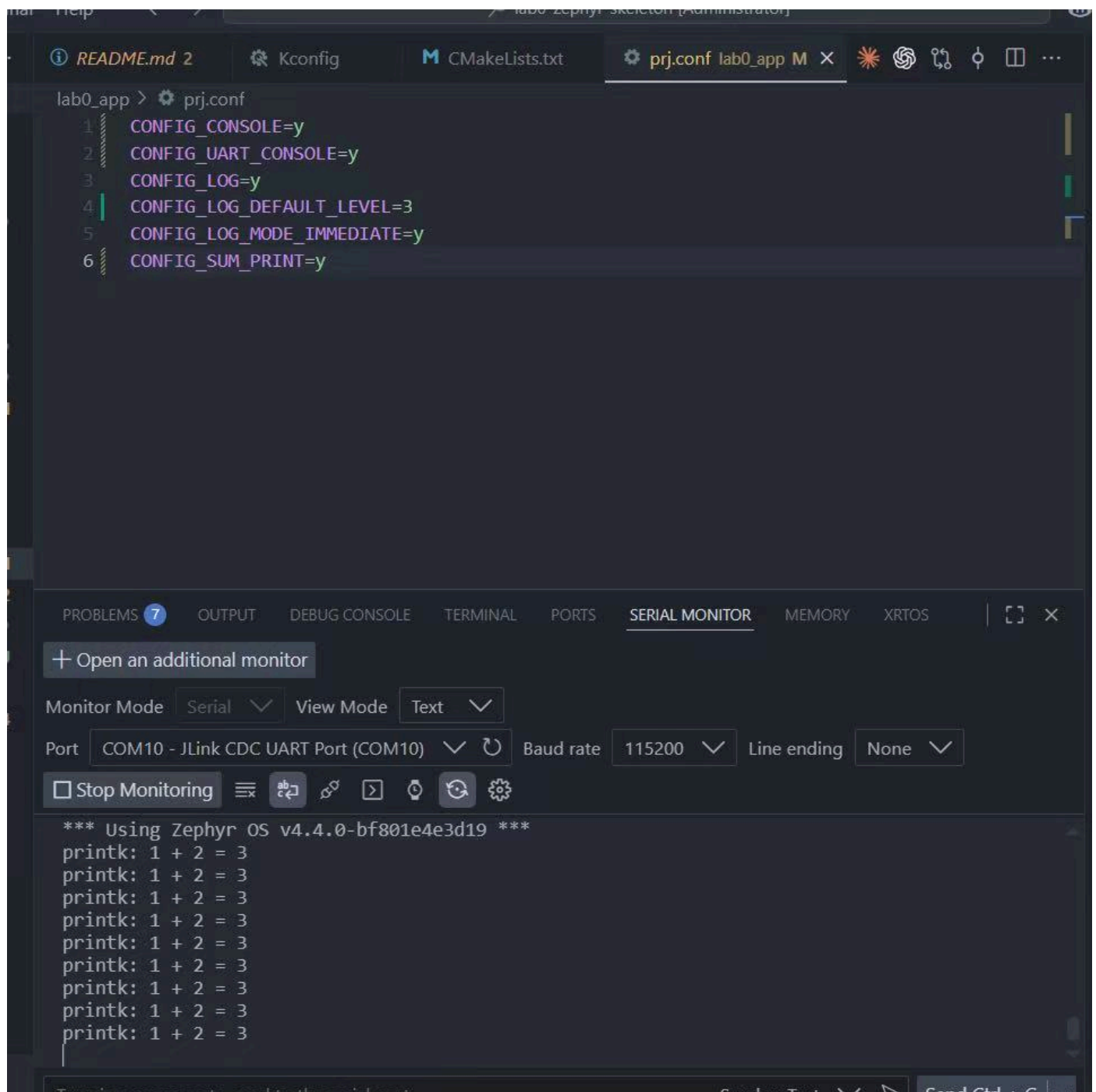
The application uses these aliases instead of directly naming the board nodes.

This keeps the C code independent from board-specific node names.

6. Printk and Logger Output

6.1 Console Output Screenshots

CONFIG_SUM_PRINT=y output with printk():



The screenshot shows the Zephyr IDE interface. The top panel displays the configuration file `prj.conf` for the `lab0_app` project. The configuration includes:

```
1 CONFIG_CONSOLE=y
2 CONFIG_UART_CONSOLE=y
3 CONFIG_LOG=y
4 CONFIG_LOG_DEFAULT_LEVEL=3
5 CONFIG_LOG_MODE_IMMEDIATE=y
6 CONFIG_SUM_PRINT=y
```

The bottom panel shows the **SERIAL MONITOR** tab. The monitor is configured to use `COM10 - JLink CDC UART Port (COM10)` at a baud rate of `115200` with `None` line ending. The output shows the following text:

```
*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
printk: 1 + 2 = 3
printk: 1 + 2 = 3
printk: 1 + 2 = 3
printk: 1 + 2 = 3
printk: 1 + 2 = 3
printk: 1 + 2 = 3
printk: 1 + 2 = 3
printk: 1 + 2 = 3
printk: 1 + 2 = 3
printk: 1 + 2 = 3
```

CONFIG_SUM_LOG=y output with the logger and hexdump:

```
lab0_app > prj.conf
1 CONFIG_CONSOLE=y
2 CONFIG_UART_CONSOLE=y
3 CONFIG_LOG=y
4 CONFIG_LOG_DEFAULT_LEVEL=3
5 CONFIG_LOG_MODE_IMMEDIATE=y
6 # CONFIG_SUM_PRINT=y
7 CONFIG_SUM_LOG=y
8
```

PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL PORTS SERIAL MONITOR MEMORY XRTOS

+ Open an additional monitor

Monitor Mode Serial View Mode Text

Port COM10 - JLink CDC UART Port (COM10) Baud rate 115200 Line ending None

Stop Monitoring

```
[00:01:58.321,411] <wrn> sum_log: warning example: computing sum with logger
[00:01:58.329,132] <inf> sum_log: sum inputs
                        01 00 00 00 02 00 00 00
.
[00:01:58.342,712] <inf> sum_log: 1 + 2 = 3
[00:02:00.347,412] <inf> sum_log: sum log implementation selected
[00:02:00.353,851] <wrn> sum_log: warning example: computing sum with logger
[00:02:00.361,541] <inf> sum_log: sum inputs
                        01 00 00 00 02 00 00 00
.
[00:02:00.375,152] <inf> sum_log: 1 + 2 = 3
[00:02:02.379,852] <inf> sum_log: sum log implementation selected
[00:02:02.386,291] <wrn> sum_log: warning example: computing sum with logger
[00:02:02.394,012] <inf> sum_log: sum inputs
                        01 00 00 00 02 00 00 00
.
[00:02:02.407,623] <inf> sum_log: 1 + 2 = 3
```

Video for Section 6.2 is submitted separately through the course Google Form.

7. Ztest for Unit Testing

QEMU Ztest output:

```

C:\Program Files\qemu\qemu-system-arm.exe: warning: nic stellaris_enet.0 has no peer
*** Booting nRF Connect SDK v3.4.0-99553055607b ***
*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
Running TESTSUITE sum_tests
=====
START - test_sum_negative

    Assertion succeeded at CMAKE_SOURCE_DIR/src/main.c:11 (sum_tests_test_sum_negative)
PASS - test_sum_negative in 0.001 seconds
=====
START - test_sum_positive

    Assertion succeeded at CMAKE_SOURCE_DIR/src/main.c:6 (sum_tests_test_sum_positive)
PASS - test_sum_positive in 0.001 seconds
=====
START - test_sum_zero

    Assertion succeeded at CMAKE_SOURCE_DIR/src/main.c:16 (sum_tests_test_sum_zero)
PASS - test_sum_zero in 0.001 seconds
=====
TESTSUITE sum_tests succeeded

----- TESTSUITE SUMMARY START -----

SUITE PASS - 100.00% [sum_tests]: pass = 3, fail = 0, skip = 0, total = 3 duration = 0.003 seconds
- PASS - [sum_tests.test_sum_negative] duration = 0.001 seconds
- PASS - [sum_tests.test_sum_positive] duration = 0.001 seconds
- PASS - [sum_tests.test_sum_zero] duration = 0.001 seconds

----- TESTSUITE SUMMARY END -----

=====
PROJECT EXECUTION SUCCESSFUL

```

Twister output:

```

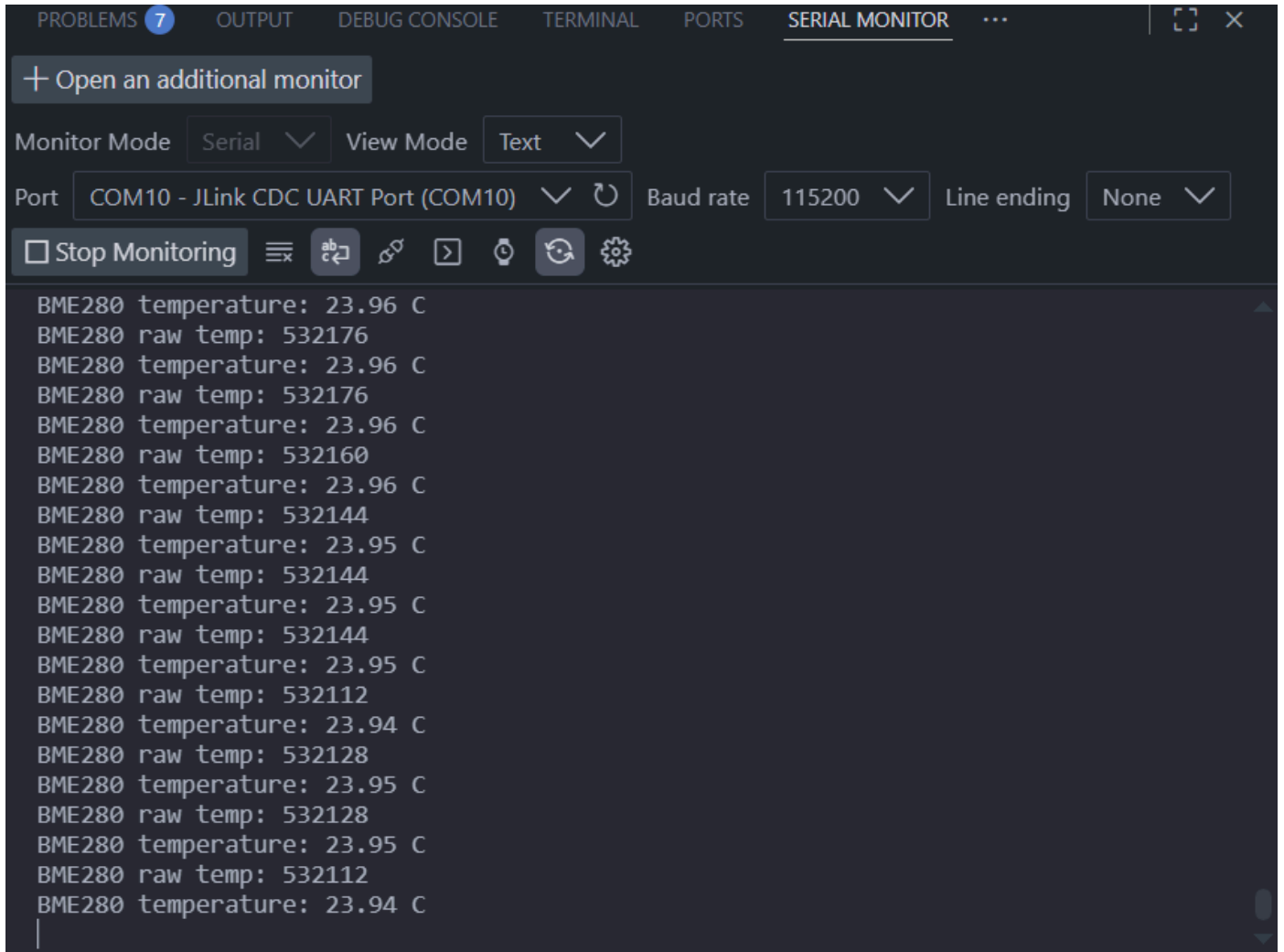
INFO    - Run completed
PS D:\ncs\v3.4.0\zephyr> west twister -T D:\ncs\lab0-tests-real\SUM_UNIT_TEST -p qemu_cortex_m3 --
inline-logs -v --clobber-output
Deleting output directory D:\ncs\v3.4.0\zephyr\twister-out
INFO    - Using Ninja..
INFO    - Zephyr version: ncs-v3.4.0
INFO    - Using 'zephyr/gnu' toolchain variant.
INFO    - Building initial testsuite list...
INFO    - Built testsuite list in 0.00 seconds
INFO    - Writing JSON report D:\ncs\v3.4.0\zephyr\twister-out\testplan.json
INFO    - Adding tasks to the queue...
INFO    - Added initial list of jobs to queue
INFO    - 1/1 qemu_cortex_m3/ti_lm3s6965 lab0.sum PA
SSED (qemu 9.579s <zephyr/gnu>)

INFO    - 1 test scenarios (1 configurations) selected, 0 configurations filtered (0 by static fil
ter, 0 at runtime).
INFO    - 1 of 1 executed test configurations passed (100.00%), 0 built (not run), 0 fai
led, 0 errored, with no warnings in 44.79 seconds.
INFO    - 3 of 3 executed test cases passed (100.00%) on 1 out of total 1514 platforms (0.07%).
INFO    - 1 test configurations executed on platforms, 0 test configurations w
ere only built.
INFO    - Saving reports...
INFO    - Writing JSON report D:\ncs\v3.4.0\zephyr\twister-out\twister.json
INFO    - Writing xunit report D:\ncs\v3.4.0\zephyr\twister-out\twister.xml...
INFO    - Writing xunit report D:\ncs\v3.4.0\zephyr\twister-out\twister_report.xml...
INFO    - Run completed
PS D:\ncs\v3.4.0\zephyr> 

```

8. Adding a BME280 Peripheral

8.1 Temperature Output



The screenshot shows the Serial Monitor window in an IDE. The top bar includes tabs for PROBLEMS (7), OUTPUT, DEBUG CONSOLE, TERMINAL, PORTS, and SERIAL MONITOR. Below the tabs is a toolbar with a button to '+ Open an additional monitor'. The 'Monitor Mode' is set to 'Serial' and 'View Mode' is set to 'Text'. The 'Port' is 'COM10 - JLink CDC UART Port (COM10)' and the 'Baud rate' is '115200'. The 'Line ending' is set to 'None'. A 'Stop Monitoring' button is on the left. The main area displays the following output:

```
BME280 temperature: 23.96 C
BME280 raw temp: 532176
BME280 temperature: 23.96 C
BME280 raw temp: 532176
BME280 temperature: 23.96 C
BME280 raw temp: 532160
BME280 temperature: 23.96 C
BME280 raw temp: 532144
BME280 temperature: 23.95 C
BME280 raw temp: 532144
BME280 temperature: 23.95 C
BME280 raw temp: 532144
BME280 temperature: 23.95 C
BME280 raw temp: 532112
BME280 temperature: 23.94 C
BME280 raw temp: 532128
BME280 temperature: 23.95 C
BME280 raw temp: 532128
BME280 temperature: 23.95 C
BME280 raw temp: 532112
BME280 temperature: 23.94 C
```

8.2 BME280 Devicetree Ztest

QEMU Ztest output:

9. Checkoff
